

$$H_0: \mu = 2400$$

$$H_1: \mu \neq 2400$$

LCL $\rightarrow$ lower

UCL $\rightarrow$ upper

standard error $\rightarrow \sqrt{\frac{s^2}{n}}$

$$Z = \frac{\bar{X} - \mu}{\sqrt{\frac{s^2}{n}}}$$

النتيجة

$\Rightarrow$ Point estimate for mean = 2489.1066 = $\bar{X}$

$$\text{Confidence interval} = \bar{X} \pm Z * \frac{\sigma}{\sqrt{n}} \Rightarrow P[2485.1867 < \mu < 2493.026] = 95\%$$

LCL

UCL

two-tail

right-tail

left-tail

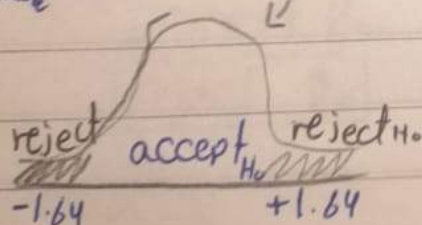
$$H_0: \mu = 2400$$

$$H_1: \mu \neq 2400$$

$$\alpha: 5\%$$

$$Z_{\text{calc}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = 44.55$$

$$Z_{\text{table}} = 1.96$$



reject H_0

accept H_1

$$p\text{-value} = 2 \times P[Z \geq 44.55] = 0.00$$

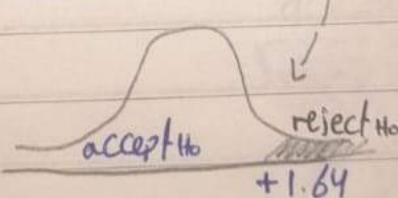
$$H_0: \mu = 2400$$

$$H_1: \mu > 2400$$

$$\alpha: 5\%$$

$$Z_{\text{calc}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = 44.55$$

$$Z_{\text{table}} = 1.6449$$



reject H_0

accept H_1

$$p\text{-value} = P[Z \geq 44.55] = 0.000$$

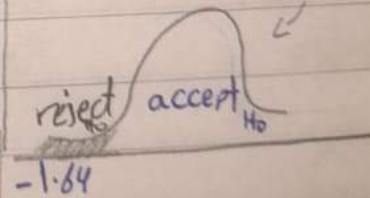
$$H_0: \mu = 2400$$

$$H_1: \mu < 2400$$

$$\alpha: 5\%$$

$$Z_{\text{calc}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = 44.55$$

$$Z_{\text{table}} = 1.6449$$



accept H_0

reject H_1

$$p\text{-value} = P[Z \leq 44.55] = 1.000$$

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لوفى Variance. متاع ال population و متاع ال variance هنا خد ال population

* independent - paired - dependent choose

* Z - T

population Variance (Known) ^{عنى}

* 7.5286 is: $\bar{X} - \boxed{\bar{X}_1 - \bar{X}_2}$ - point estimate for mean - point estimate for difference mean

* 3.7670 is: Lower confidence interval - $\bar{X} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ - all for difference mean

* Confidence interval:

$$P[11.29 < \mu_1 - \mu_2 < 3.76] = 95\%$$

$$P[3.76 < \bar{X}_1 - \bar{X}_2 < 11.29] = 95\%$$

$$\boxed{P[3.76 < \mu_1 - \mu_2 < 11.29] = 95\%}$$

* Hypo is:

$$H_0: \mu_1 - \mu_2 = 7.5286, H_1: \mu_1 - \mu_2 \neq 7.5286 \quad \times$$

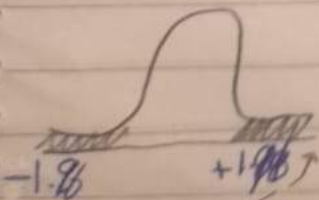
$$H_0: \mu_1 - \mu_2 \neq \text{zero}, H_1: \mu_1 - \mu_2 = \text{zero} \quad \times$$

$$\boxed{H_0: \mu_1 - \mu_2 = \text{zero}, H_1: \mu_1 - \mu_2 \neq \text{zero}}$$

* two-tail:

accept H_0

reject H_0

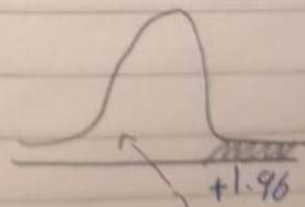


$$Z = 3.92$$

* right tail:

accept H_0

reject H_0

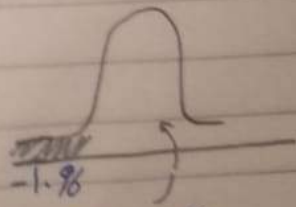


$$Z = 3.92$$

* left tail:

accept H_0

reject H_0



$$Z = 3.92$$

page 4:

* paired - independent* T - Z
↓
unknown ← population variance sample* mean difference: $\bar{X}_1 - \bar{X}_2$ - point estimate - all

pooled variance:
$$S_p^2 = \frac{S_1^2 [n_1 - 1] + S_2^2 [n_2 - 1]}{n_1 + n_2 - 2}$$

population variance is
unknown but equal.

* $df = n - 1$

- $df = n_1 + n_2 - 2$

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* paired - independent

المجموعة الأولى $\rightarrow P[40.85 < \mu_1 < 68.37] = 95\%$

المجموعة الثانية $\rightarrow P[9.7 < \mu_2 < 34.02] = 95\%$

الفرق $\rightarrow P[15.37 < \mu_1 - \mu_2 < 42.10] = 95\%$

* Hypothesis difference $\Rightarrow H_0: \mu_1 - \mu_2 = \text{zero}$
 $H_1: \mu_1 - \mu_2 \neq \text{zero}$

* point estimate: $\bar{X}_1 - \bar{X}_2 = 30.24$

* degree of freedom: $n-1$

$$\frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$\frac{n_1 + n_2 - 2}{\frac{n_1}{n_1 - 1} + \frac{n_2}{n_2 - 1}}$$

مساكني التفاضل

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* Paired - independent
↓
= dependent

before $\Rightarrow P[10.35 < \mu_1 < 19.07] = 95\%$

after $\Rightarrow P[16.34 < \mu_2 < 23.07] = 95\%$

1-2 $\Rightarrow P[3 < \mu_d < 6.99] = 95\%$

~~μ_1, μ_2~~ $\rightarrow$ so independent

* mean difference: $\bar{X}_1 - \bar{X}_2 = -5$ — $\bar{X}_d = -5$

* degree of freedom: $n-1$
 $n_1 + n_2 - 2$

*

$H_0: \mu_1 - \mu_2 = \text{zero}$

$H_1: \mu_1 - \mu_2 \neq \text{zero}$

independent ↙

$H_0: \mu_d = \text{zero}$

$H_1: \mu_d \neq \text{zero}$

paired ↙